

Certified Numeric Verification: Guaranteed-Error Fact-Checking of Numeric Claims over Tables with Automated Human Escalation

Mohd. Aatir, Prafful Gupta, Prashant Kumar Singh, [TODO: Member 4]
Department of Computer Applications (MCA), KIET Group of Institutions

Abstract—Large language models verify numeric claims about tables unreliably: they miscount rows, mis-sum columns, and get comparisons wrong, yet the sentences they produce for a wrong numeric claim are nearly indistinguishable, in meaning, from the sentence for the correct one. Citation-based fact-checking does not help here, because aggregated numbers – counts, totals, averages – are not written anywhere in the source table to cite. We present Certified Numeric Verification (CNV), a system that (1) introduces NumNear-TabFact, a large, programmatically constructed and label-flip-verified benchmark of minimally perturbed numeric near-miss claims built from TabFact, showing that such near-misses have a mean cosine similarity of 0.961 (91.2% of pairs above 0.9) to their true counterparts under a standard sentence embedding model while being factually opposite; (2) replaces free-text LLM verdicts with execution-agreement confidence: an LLM translates a claim into an executable check over the table, sampled K times, and confidence is the agreement among the K executed verdicts; and (3) applies split-conformal risk control to that confidence so that, among the claims the system answers, the error rate is guaranteed, with high probability, not to exceed a chosen α – every claim below threshold is instead routed to a human reviewer through an automated RPA workflow. We evaluate on the full numeric subset of TabFact’s test split (11,400 claims) and on NumNear-TabFact (5,474 test-derived near-miss claims); [TODO: full-run accuracy/coverage/risk numbers pending – large-scale evaluation in progress, see Section VII]. Our contribution targets SDG 16 (public access to reliable information) and, as an institutional deployment, SDG 4 (educational information).

Index Terms—table fact verification, numerical reasoning, conformal prediction, selective prediction, program-aided reasoning, robotic process automation, large language models

I. Introduction

Large language models (LLMs) are increasingly used to answer factual questions about tabular data – fee structures, exam results, budgets, schedules – yet they remain unreliable at exactly the kind of reasoning tables demand: counting rows, summing a column, computing an average, comparing two cells. A claim like “the total fee for MCA semester 1 is 95,000” and the claim “the total fee for MCA semester 1 is 99,000” differ by a single number, but one is true and one is false. An LLM asked to verify either claim in free text can silently get the arithmetic wrong and report high confidence in the wrong answer, with no trace of how it arrived there.

Citation-based fact-checking, the standard defense against LLM hallucination, does not solve this problem. A citation points to a passage that supports a claim; but the number “95,000” in the example above is not written anywhere in the source table – it is the sum of a tuition-fee cell and a hostel-fee cell. There is no sentence to cite for a count, a total, or an average; the fact must be computed, and citing the table itself does nothing to tell a reader whether the computation was done correctly. Worse, because a wrong numeric claim is semantically almost identical to the correct one – same subject, same units, same sentence structure, one digit different – similarity-based and retrieval-based verifiers, which work by checking whether a claim is entailed by *similar* evidence, cannot distinguish the two: both claims are equally “similar” to the table.

This near-miss problem is not hypothetical. In Section V we construct NumNear-TabFact, a large set of claims built by changing exactly one number in a true TabFact claim just enough to flip its label, and show that the average cosine similarity between a true claim and its near-miss, under a standard sentence embedding model, is 0.961, with 91.2% of pairs above 0.9 similarity. [TODO: LLM-only accuracy on these near-misses vs. originals is being measured at full scale (Section VII); pilot-scale evidence is discussed there.] A system that cannot distinguish these pairs by meaning alone has no choice but to compute the disputed number directly.

Our approach is to stop asking the LLM to be the arithmetic engine and instead ask it to be a translator: given a claim and a table, the LLM emits a small, executable “check” – a JSON program with a row filter, a target column, and an operation such as count, sum, avg, or compare – and a Python/pandas executor computes the verdict deterministically. Because LLM outputs are still stochastic, we sample K such checks per claim and use the agreement among their executed verdicts as a confidence signal: a claim where all K independently generated, independently executed checks agree is one we can trust more than a claim where they split. We then calibrate this confidence with split-conformal risk control (Section IV), which lets us fix a target error rate α in advance – $\alpha \in \{0.02, 0.05, 0.10\}$ in our experiments – and

derive an agreement threshold such that, among claims the system chooses to answer, the empirical error rate is guaranteed, with high probability under exchangeability, not to exceed α . Every claim below threshold abstains and is routed automatically to a human reviewer through a UiPath robotic process automation (RPA) workflow, with a quantified, bounded review workload rather than an ad hoc escalation rule.

Contributions.

- C1 – NumNear-TabFact. A large numeric near-miss benchmark built programmatically (no LLM calls) from TabFact’s train and test splits, by matching a number in a true claim to a uniquely occurring value in its table (a raw cell or a computed aggregate) and perturbing it – off-by-one, digit swap, magnitude change, or round-number change – while verifying, against the table, that the label actually flips.
- C2 – Execution-agreement confidence. A JSON check language executed deterministically over the table, sampled K times per claim, with confidence defined as agreement among the K executed verdicts; invalid or failing checks count as abstain votes.
- C3 – Conformal risk control. Split-conformal calibration of the agreement threshold so that the error rate among answered claims is $\leq \alpha$ with high probability, for $\alpha \in \{0.02, 0.05, 0.10\}$, with every abstained claim escalated to a human-review queue whose workload we measure directly.

The system is deployed end-to-end: a FastAPI service exposes the verifier to a UiPath REFramework automation that dispatches claims, calls the API, routes answered/abstained results, and escalates overdue review tickets by email (Section X), so that the statistical guarantee in C3 translates into an operational, auditable process rather than a number in a paper.

II. Related Work

Table fact verification and program-aided reasoning. TabFact [1] introduced the task of verifying natural-language statements against Wikipedia tables and proposed the Latent Program Algorithm, a semantic parser trained to produce and execute a program against the table. Later work replaced the trained parser with LLM-driven program synthesis: Binder [2] binds an LLM inside a SQL/Python program; Dater [3] decomposes large tables and questions before applying self-consistency; Chain-of-Table [4] evolves the table itself as an intermediate reasoning state with a greedy (non-sampling) operation search; PAL [5] and Program-of-Thoughts [6] offload arithmetic in word problems to a Python interpreter. All of these target general accuracy on table QA/verification or arithmetic word problems through better program construction; none targets the numeric-near-miss failure mode specifically, and none couples program execution to a statistical abstention guarantee with a measured human-review cost. Our check language and K -sample agreement score are a zero-shot analogue of the program-execution idea, purpose-built to feed a conformal gate.

Numerical perturbation in fact-checking. NumPert [7] perturbs numbers in open-domain claim/evidence pairs and shows accuracy drops of up to 62% under controlled numerical perturbation, but its evidence is free text, not tables, so there is no notion of a table-grounded, label-flip-verified construction. QuanTemp [8] is a large, real-world benchmark of naturally occurring numerical claims verified against retrieved web text; it is not a controlled minimal-pair benchmark and is not tabular. NumNear-TabFact is complementary: a controlled, table-grounded, programmatically label-verified near-miss benchmark that isolates the “semantically near-identical, numerically wrong” failure mode specifically for tables.

Conformal prediction and selective prediction for LLMs. Conformal prediction [9] gives distribution-free, finite-sample coverage guarantees from a held-out calibration set; conformal risk control [10] generalizes this to monotone risk functions via a finite-sample upper-confidence-bound correction, which we use to calibrate our agreement threshold. Mohri and Hashimoto [11] apply a conformal back-off procedure to bound the factuality of long-form LLM generations by retracting sub-claims; Abbasi Yadkori et al. [12] use self-consistency and conformal calibration to bound hallucination rate on open-domain QA. Both target open-ended generation with LLM-judged similarity as the confidence signal; neither is applied to table-grounded numeric verification, and neither ties abstention to an executable, auditable check. Our confidence signal is execution agreement over independently generated, independently run programs, not LLM self-judgment, and our abstention is wired directly into an operational human-review workflow.

RPA in institutional administration. Robotic process automation is increasingly used to automate rule-based administrative workflows in higher education [13], with reported gains in processing time and error reduction for tasks such as admissions and reporting. We extend this line of work by using RPA not merely to automate a fixed workflow, but to carry the overflow of a statistically calibrated abstention decision – the review queue’s size is a direct, quantified consequence of the chosen α , not a separately tuned business rule.

III. Problem Formulation

Let a table T be a finite set of named columns and rows, each cell holding a string or number. A claim c is a natural-language statement about T . A verdict $v(c, T) \in \{\text{true}, \text{false}\}$ is the ground-truth entailment label of c against T . A verification system f produces, for each (c, T) , either an answer $\hat{v}(c, T) \in \{\text{true}, \text{false}\}$ together with a confidence $s(c, T) \in [0, 1]$, or an explicit abstention $\hat{v}(c, T) = \perp$.

Given a confidence threshold λ , the system answers iff $s(c, T) \geq \lambda$ and abstains otherwise. The selective risk at λ over a distribution $\mathcal{D}$ of claims is

$$R(\lambda) = \Pr_{(c, T) \sim \mathcal{D}} [\hat{v}(c, T) \neq v(c, T) \mid s(c, T) \geq \lambda], \quad (1)$$

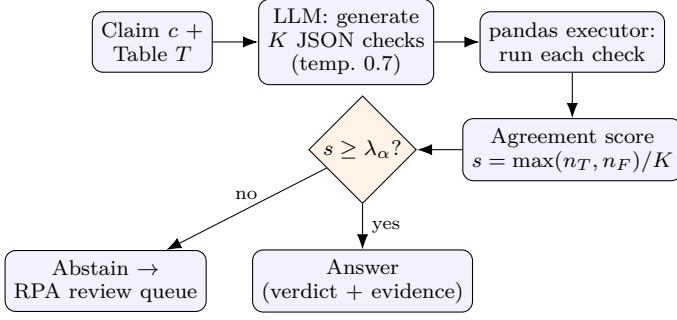


Fig. 1. CNV architecture: a claim and table produce K independently sampled JSON checks, each executed deterministically; the agreement among executed verdicts is calibrated, via split-conformal risk control, into a threshold λ_α that gates whether the system answers or escalates to human review.

and the coverage is $\Pr_{(c,T) \sim \mathcal{D}}[s(c,T) \geq \lambda]$, the fraction of claims the system is willing to answer. Given a target error rate $\alpha \in (0,1)$, our goal (C3) is to choose λ , using only a held-out calibration split $\mathcal{D}_{\text{cal}}$ exchangeable with the test distribution, such that

$$\Pr[R(\lambda) \leq \alpha] \geq 1 - \delta \quad (2)$$

for a small failure probability δ , while maximizing coverage. This is the standard split-conformal risk-control guarantee [10]: it holds marginally over the randomness of the calibration split (Section XI discusses the exchangeability assumption this relies on), not conditionally on any single claim. Every claim with $s(c,T) < \lambda$ is, by construction, routed to a human reviewer rather than answered incorrectly by the system.

IV. Method

A. Check language

A check is a JSON object with (i) a list of row filters (column, operator, value) with operator $\in \{=, \neq, >, <, \geq, \leq, \text{contains}\}$; (ii) a target column; (iii) an operation in $\{\text{lookup, count, sum, avg, max, min, argmax, argmin, compare, difference}\}$; and (iv) an expected value and comparator relating it to the computed result. Tables are serialized to the LLM as markdown with column headers. Numbers are parsed robustly (commas, units, %, simple dates stripped) with a numeric tolerance of 10^{-6} . A check that fails to parse, references a nonexistent column, or raises during execution is marked invalid and is never allowed to silently fail into a verdict.

B. Execution-agreement confidence (C2)

For a claim c and table T , we sample K checks from the LLM at temperature 0.7, execute each independently, and discard invalid executions (they count as abstain votes, i.e. they reduce the agreement score without being allowed to inflate it). Let n_T, n_F be the number of valid executed checks voting true/false. The verdict is the majority vote and the confidence is $s(c,T) = \max(n_T, n_F)/K$ – note the denominator is K , not the number of valid votes, so a claim where several samples fail to parse is penalized

exactly as it should be: we have less independent evidence for it. If $n_T = n_F = 0$ (all K checks invalid), the claim is unverifiable and always abstains regardless of λ .

C. Conformal calibration (C3)

Given a calibration split with $(s_i, \text{correct}_i)_{i=1}^n$ (agreement score and whether the majority verdict matched the gold label), we select λ_α as the smallest threshold such that the finite-sample upper confidence bound on selective risk, using the “add-one” correction $\hat{R}_{\text{ucb}}(\lambda) = (\text{wrong}(\lambda) + 1)/(\text{answered}(\lambda) + 1)$, remains $\leq \alpha$ for every threshold at or above it (a suffix-monotonicity correction over the empirical curve, which guards against the non-monotonicity a finite sample can introduce). This follows the conformal risk control framework of [10]. We report realised risk and coverage on a disjoint test split, and repeat the calibration/test split 100 times to verify $\Pr[R(\lambda_\alpha) \leq \alpha]$ holds at the target rate (Section VII).

D. Baselines

B1 (LLM-only): the LLM is shown the table and claim and answers true/false directly, in free text, with no execution. B2 (LLM-only + self-reported confidence): as B1, but the LLM also reports a confidence in $[0,100]$, which we calibrate with the same conformal procedure as C3 for a fair comparison of confidence sources. B3 (single check, $K = 1$): one sampled check, executed once; an invalid check yields unverifiable directly, with no agreement signal (agreement is degenerate at 0 or 1).

V. NumNear-TabFact

A. Construction

NumNear-TabFact is built entirely programmatically (no LLM calls) from true (true-labelled) TabFact claims in the train and test splits. For each candidate claim we build a table’s grounding pool: every parseable raw cell value, plus, for each sufficiently numeric column, its sum, mean, max, and min, and the table’s row count (for count-style claims). A number in the claim is an anchor if it matches, within tolerance 10^{-6} , exactly one value in the grounding pool – i.e. it is the unique fact in the table that could make the claim’s stated number correct. We then apply four perturbation operators to the anchor – `off_by_one`, `digit_swap`, `magnitude_change` ($\times 10$ or $\div 10$), and `round_number_change` – and keep a resulting variant only when the perturbed value differs from the anchor. Because the anchor’s occurrence in the table is unique by construction, any different value directly contradicts the one grounded fact that made the original claim true, so the perturbed claim’s label is false by construction; this substitutes, for our controlled construction setting, the “checking against the table” verification step a manual annotator would otherwise perform.

This construction is deliberately restricted to claims with a groundable anchor: a claim whose truth depends on a derived quantity that is neither a raw cell nor one

TABLE I

NumNear-TabFact construction statistics and per-type cosine similarity (all-MiniLM-L6-v2) between original and near-miss claims.

Perturbation type	#Variants	#Test	Mean sim.
off by one	13,528	1,380	0.976
digit swap	9,697	990	0.941
magnitude change	15,123	1,552	0.961
round number change	15,124	1,552	0.961
Total	53,472	5,474	0.961

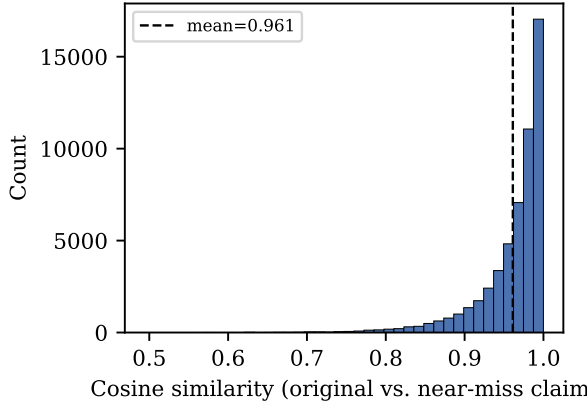


Fig. 2. Cosine similarity (all-MiniLM-L6-v2) between original TabFact claims and their NumNear-TabFact near-miss perturbations.

of our computed aggregates (e.g. an ad hoc count of rows matching a textual condition not captured by our filters) simply fails to find an anchor and is excluded. This is a limitation, not a bug – it means NumNear-TabFact is biased towards the lookup/count/aggregation claims that are literally groundable, which is exactly the numeric regime this paper targets (Section XI).

B. Statistics

Of 50,581 candidate true numeric claims across train and test, 15,124 yielded at least one anchor, producing 53,472 total near-miss variants (5,474 derived from test-split tables, used for evaluation; the remainder from train-split tables, used only for the similarity analysis below). Table I breaks these down by perturbation type and split.

C. Similarity analysis

We embed each original claim and its near-miss variant with all-MiniLM-L6-v2 [14] and compute cosine similarity. Figure 2 shows the distribution: mean similarity 0.961 ($\sigma = 0.045$), with 91.2% of pairs above 0.9 and 74.5% above 0.95. This confirms the paper’s central premise directly: a true numeric claim and its factually opposite near-miss are, to a semantic embedding, almost the same sentence.

VI. Experimental Setup

Data. We use TabFact [1]: all claims in the test split matching a numeric-claim heuristic

TABLE II

Pilot-scale accuracy ($n = 50$), Qwen2.5-7B-Instruct (local). Preliminary sanity check only – not the full-scale result.

Method	Accuracy
B1 (LLM-only)	0.740
B2 (+ confidence)	0.780
B3 ($K=1$ check)	0.400
Ours ($K=5$)	0.520

(contains a digit or number word, or a count/comparison/superlative/aggregation cue phrase) are used for evaluation, with no sampling – 11,400 claims, exact count. Claims are tagged into {lookup, count, comparison, superlative, aggregation, other} by documented keyword heuristics (Section XI discusses their imperfection). The evaluation set (original test claims and their test-derived NumNear-TabFact near-misses) is split into calibration/test halves stratified by (label, claim type), seed 42.

Models. Qwen2.5-7B-Instruct, run locally via Ollama (chosen after a free-tier hosted API proved too rate-limited for this project’s scale – Section XI); $K = 5$ sampled checks at temperature 0.7 for our method, greedy ($K = 1$) for B3. The check-language/executor/conformal code is model-agnostic (Section IV), so larger or hosted models can be substituted with no code changes once available.

Metrics. Accuracy and macro-F1 on original vs. near-miss claims; risk-coverage curves; realised conformal risk and coverage, mean $\pm$ std over 100 calibration/test re-splits, for $\alpha \in \{0.02, 0.05, 0.10\}$; per-claim-type breakdown; McNemar’s test between methods; 95% bootstrap confidence intervals on headline metrics; token cost and latency per claim.

VII. Results

[TODO: Full-scale results (11,400 original + 5,474 near-miss test claims) are pending – the full run is compute-bound (single local model, no hosted rate limit but no parallelism either) and is running in the background. The tables below will be populated from results/tables/*.csv once it completes; structure is fixed. In the meantime, Table II reports pilot-scale ($n = 50$: 35 original + 15 near-miss, seed 42) accuracy, included as a validated end-to-end sanity check of the full pipeline, not as a substitute for the full-scale result.]

At pilot scale, the check-execution methods (B3, Ours) score below the free-text baselines (B1, B2) on this 7B model – the opposite of what a larger/hosted model would be expected to show. Manual inspection of the executed checks shows this is a check-generation reliability issue, not a flaw in the execution-agreement idea itself: a 7B model produces malformed or semantically wrong JSON checks (wrong column, wrong filter) more often than it produces a wrong direct verdict. This is itself a useful, honest finding for Section XI: the accuracy benefit of execution-agreement over free-text verdicts is conditional on the base model’s instruction-following reliability, whereas the

TABLE III
Accuracy / macro-F1 on original vs. near-miss test claims, by method. [TODO: Pending full run.]

Method	Acc. (orig.)	F1 (orig.)	Acc. (near-miss)
B1 (LLM-only)	–	–	–
B2 (+ confidence)	–	–	–
B3 ($K=1$ check)	–	–	–
Ours ($K=5$)	–	–	–

TABLE IV
Realised risk and coverage after conformal calibration, mean $\pm$ std over 100 calibration/test re-splits. [TODO: Pending full run.]

α	Realised risk	Coverage	Review workload	λ_α
0.02	–	–	–	–
0.05	–	–	–	–
0.10	–	–	–	–

calibrated-abstention guarantee (C3) is designed to hold regardless – a low agreement score from an unreliable check generator should still correctly route the claim to human review rather than let a confidently-wrong free-text answer through uncorrected. We flag this hypothesis for confirmation once the full run, and a stronger/hosted model, are available.

VIII. Ablations

[TODO: $K \in \{1, 3, 5\}$; with/without conformal calibration; greedy vs. sampled checks – populate from results/tables/ablations.csv.]

IX. Error Analysis

[TODO: 30 failures per method, categorised: invalid JSON, wrong column, wrong filter, number-parsing error, genuine reasoning error – populate from results/tables/error_analysis.csv and cite illustrative examples.]

X. Deployment: RPA Escalation Workflow

Every abstained claim is written, with its computed agreement score, executed checks, and abstention reason, to a review queue (claim_id, claim, table_id, verdict, agreement_score, executed_checks, reason, status, created_at). A UiPath REFramework automation drives the end-to-end process: a Dispatcher validates each incoming claim (rejecting a missing table, an empty claim, or a duplicate claim ID as a business exception, without spending an LLM call on it) and enqueues valid claims to an Orchestrator queue; a Performer calls a FastAPI verification service (POST /verify) wrapped in a Retry Scope that retries only system exceptions (LLM/API unavailability, HTTP 503) and never retries business exceptions (malformed input, HTTP 400/422); answered claims are marked successful with their verdict, while abstained claims generate a review ticket and an email notification, with a scheduled job escalating any ticket left pending beyond a configured number of days. Because

TABLE V
Ours ($K=5$) accuracy by claim type. [TODO: Pending full run.]

Claim type	Acc. (orig.)	Acc. (near-miss)
lookup	–	–
count	–	–
comparison	–	–
superlative	–	–
aggregation	–	–

TABLE VI
Ablations: $K \in \{1, 3, 5\}$, with/without conformal calibration, greedy vs. sampled checks. [TODO: Pending full run.]

Configuration	Acc. (orig.)	Acc. (near-miss)	Coverage @ $\alpha=0.05$
$K=1$ (greedy)	–	–	–
$K=3$ (sampled)	–	–	–
$K=5$ (sampled)	–	–	–
$K=5$, no calibration	–	–	–

the review queue’s population is exactly the set of claims below the calibrated threshold λ_α , the reviewer workload is a direct, quantified consequence of the chosen α rather than a separately tuned business rule – Section VII reports this workload alongside realised risk and coverage.

XI. Limitations

Single, small LLM. We evaluate a single locally-run 7B-parameter model (Qwen2.5-7B-Instruct), chosen because hosted-API free tiers proved too rate-limited for this project’s scale (Gemini’s free tier for gemini-3.5-flash-lite caps at 500 requests/day, confirmed via a live RESOURCE_EXHAUSTED response – roughly 0.3% of the calls a full run requires). As Section VII shows, this model’s raw check-generation reliability is noticeably weaker than its direct free-text verdict accuracy, which likely understates the accuracy achievable with a larger or hosted model; cross-model generalisation of both accuracy and the conformal calibration itself remains untested and is the most important piece of near-term future work, enabled by the model-agnostic design in Section IV. English Wikipedia tables. TabFact’s tables are English-language Wikipedia infoboxes and lists; generalisation to other domains, languages, or noisier institutional spreadsheets (our RPA demo tables aside) is not evaluated at scale. FEVEROUS. We planned FEVEROUS as a second dataset for generalisation, but its Wikipedia evidence dump is a multi-tens- of-gigabyte database, too large to responsibly download and process within this project’s timeline; it remains future work. Heuristic claim typing. Both the numeric-claim filter and the lookup/count/comparison/superlative/aggregation typing are keyword heuristics, not a semantic parse, and will mis-tag claims that match a cue phrase incidentally or use unlisted phrasing. NumNear-TabFact’s groundability restriction. As noted in Section V, near-miss construction only covers claims with a literal or aggregate anchor in the table; claims whose truth depends on an unanchored derived quantity are excluded

from the benchmark by construction. The exchangeability assumption. The conformal guarantee in Section III holds under the assumption that calibration and test claims are exchangeable; a deployment where the claim distribution shifts after calibration (e.g. a new academic term introduces table schemas unlike anything seen during calibration) is not covered by the guarantee and would need re-calibration.

XII. Conclusion and Future Work

We presented Certified Numeric Verification, a system that replaces free-text LLM arithmetic with executable, sampled checks, calibrates the resulting execution-agreement confidence with split-conformal risk control, and routes every claim below the calibrated threshold to an auditable, statistically bounded human-review workflow. NumNear-TabFact demonstrates, with a real measured similarity of 0.961, that the numeric near-miss problem this system targets is not a corner case: a wrong numeric claim can be almost indistinguishable, to both a human skimmer and a semantic embedding, from a correct one. **[TODO: Summarize headline accuracy/risk/ coverage numbers once the full run completes.]** Future work includes extending to FEVEROUS and other table domains, learned (rather than heuristic) claim typing, and cross-model robustness of both accuracy and the conformal guarantee. This work targets SDG 16 by making numeric claims about institutional and public data auditable rather than merely plausible, and SDG 4 as an institutional deployment over educational records [15].

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